

Classification or learning of an image or [Decision theoretic methods]

Object Recognition & Classification

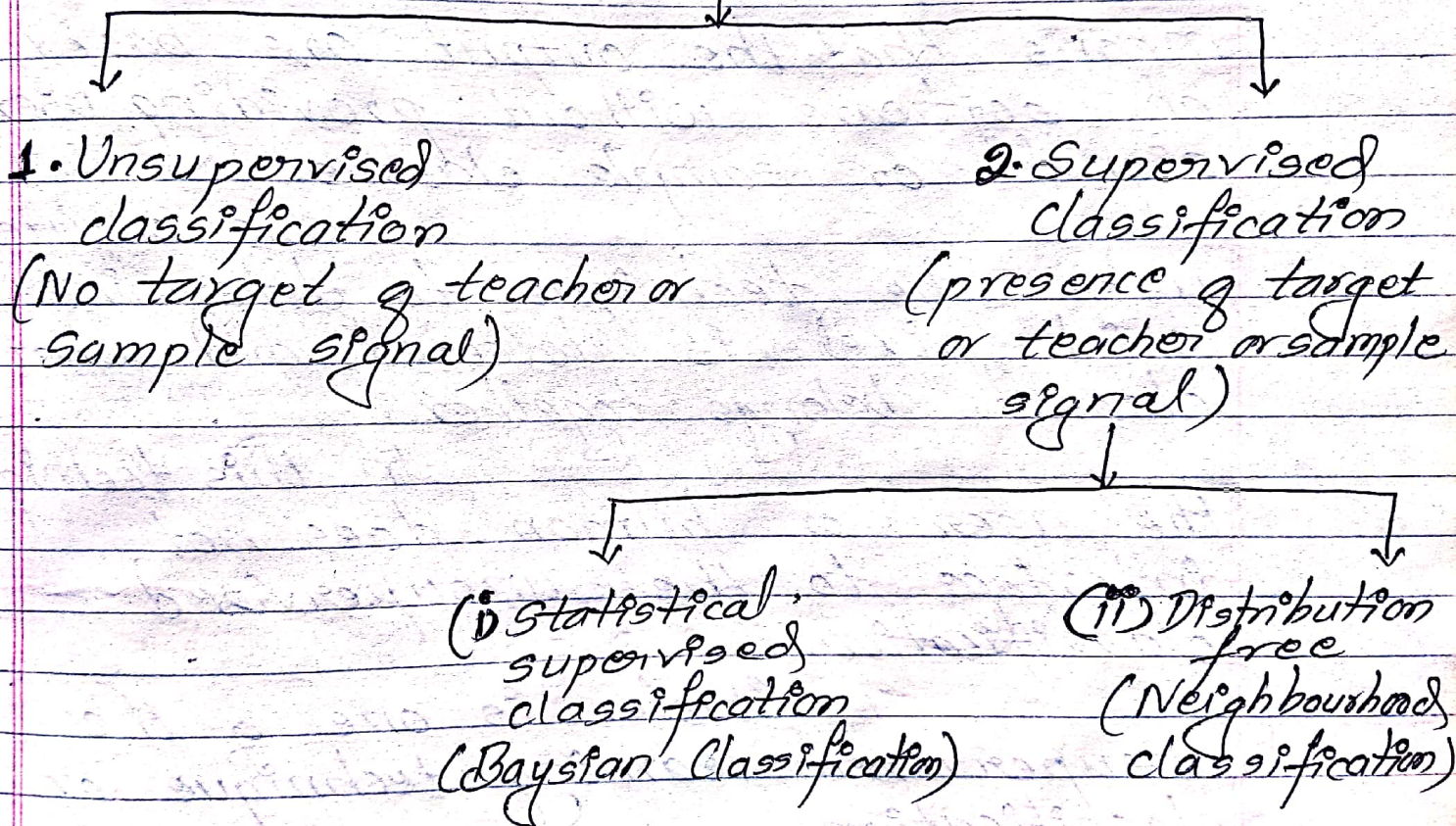


Fig: Decision Theoretic Methods

Unsupervised Classification

A very common task concerning in image processing is classification is done in order to use the image for mapping or further analysis.

In unsupervised classification learning, there is no teacher or target or sample signal. In the same way, it is also computerized method without direction of analysis. It

means that the output are based on software without providing user signal or sample classes.

The computer uses the technique to determine which pixel is detected and what classes belongs together.

In this technique the user or human does not interface to the unsupervised classification.

Cluster is one of the unsupervised learning technique or classification.

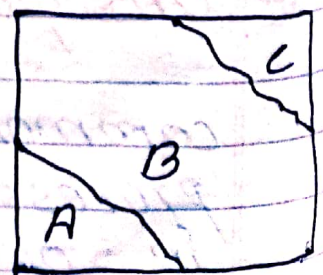
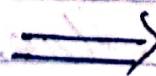
for unsupervised classification the figure below shows the technique of object recognition:

101	101	105	105
101	101	101	105
103	101	101	101
103	103	101	101

I/P Image



B	B	C	C
B	B	B	C
A	B	B	B
A	A	B	B



class classification

fig: Example of unsupervised learning using clustering

(A = water
B = land
C = rock)

03

10

Supervised Classification

In supervised classification, it does not use the computer software to create the classes.

Here, the analysis identifies several areas in an image which represents different features. These known areas are referred to as training sites.

The computer only do the assignment of pixel to the classes. It is based on the idea that a user can select sample pixel or target signal in an image i.e. representative of specific classification.

It means that the provided user signal or teacher signal can help to make the classification without using computer tool.

Supervised classification is categorized into two types:

(i) Statistical Supervised Classification

(ii) Distribution free

(04)

(i) Statistical Supervised Classification

It is a type of supervised classification on which the hypothetical or probability of that classification is determined using presence of user signal (or sample signal or teacher signal or target signal).

This still can be extended to classification of features or pattern where similar type of pattern are grouped into one classes that same as the characteristic of user signal.

The term similar in the containing objects is closeness to the features between sample signal and segmented region.

To justify this classification we use Bayesian classifier.

✓ Bayesian Classifier

A Bayesian classifier is a simple and probabilistic classifier based on the Bayesian theorem.

The Bayesian classifier classifies features to contribute the probability

(05)

of hypothetical evidence i.e.,

$$P(H_i/E) = \frac{P(E/H_i) P(H_i)}{\sum_{n=1}^K P(E/H_n) P(H_n)}$$

where,

$P(E/H_i)$ = The probability that we will observe evidence 'E' given that hypothesis H_i is true.

$P(H_i)$ = A prior probability that the hypothesis H_i is true.

K = The no. of possible hypotheses

$P(H_i/E)$ = The probability that we will observe hypothesis H_i given that hypothesis 'E' is true.

(B) Distribution free

It is a type of supervised classification & do not require knowledge of any density function on pdf & based on the heuristic information or algorithm.

The example of distribution free is nearest neighbourhood

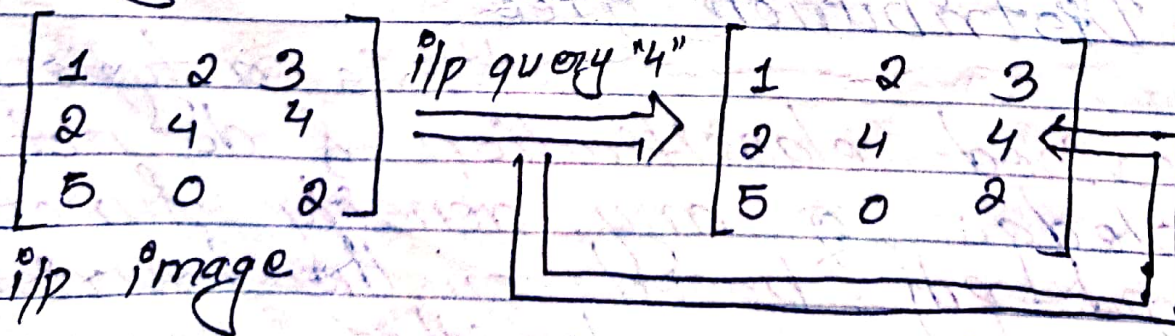
classification, so that it is also called as neighbourhood classification.

A very commonly used classification method is called nearest neighbourhood classification, therefore it works as follows:

→ We create a database of any object for which we already know that the correct classification should be known.

When the system is given a query i.e. new object classify the system. It finds the nearest neighbour of the query in the database.

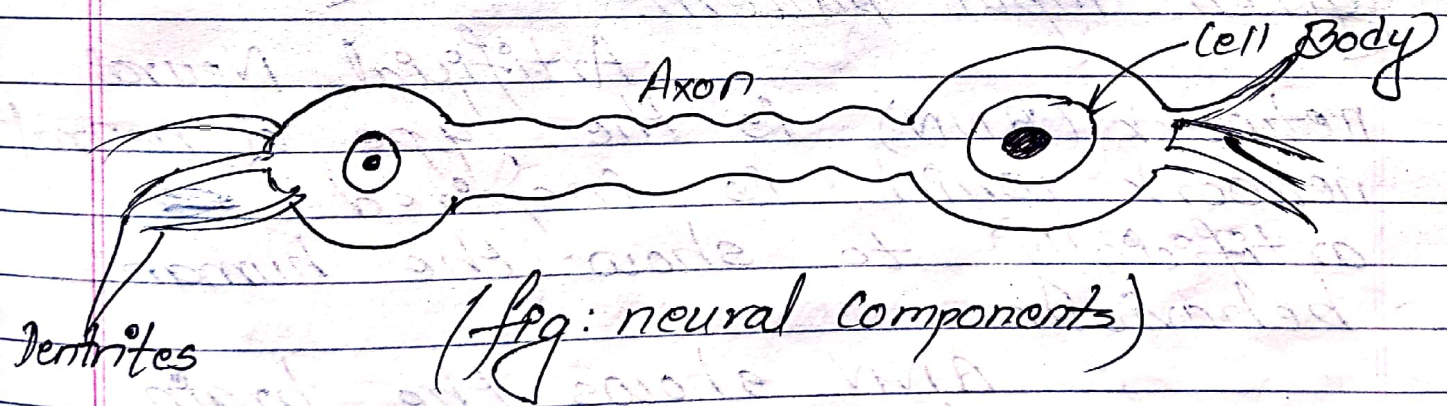
→ Then, the system classify the query as belongs to the same classes as its nearest neighbours i.e. database images.



(fig: Recognizing digit using neighbourhood classifier)

- Then, the system classify the query as image of 4.
- Now, the image is identified and classified by the distribution free supervised classification.
- Here query 4 is the target signal by the user.

Neuron, Neural Network (NN) & Artificial Neural Network (ANN)



A neuron is a cell in the brain whose principal function is the collection, processing and dissemination of electrical signal.

The cell body is the part of the cell containing the nucleus and maintaining protein synthesis.

A neuron may have

many dendrites which branch out tree like structures and receive signals from other neurons.

The axon conducts electrical signals generated at the axon hillock.

Neural Network & Artificial Neural Network

Neural network is the ^{branch} ~~branch~~ of the AI that can be considered as blackbox ~~de~~ that is able to predict the output pattern when it recognizes the given input pattern.

Artificial Neural network (ANN) is the type of neural network which is inspired ~~is~~ artificially to show the human behaviour.

ANN shows the brain process information system.

Q Why use neural network (NN)?

⇒ Neural network is only used for following:

- (a) Self organization
- (b) Real Time Operation
- (c) Information is distributed over the entire network
- (d) Adaptive Learning.

Applications of Neural Network

(i) Automotives, automobile and automatic guidance system

(ii) Aerospace and aircraft components simulation.

(iii) Banking cheque ~~return~~ reader

(iv) ~~Difference~~ ^{Diff} facial recognition

(v) Manufacturing product design (CAD)

(vi) Telecommunication Automated Information

(vii) Robotics Control System

Disadvantages of NN

(i) NN needs training to operate & lack of manpower.

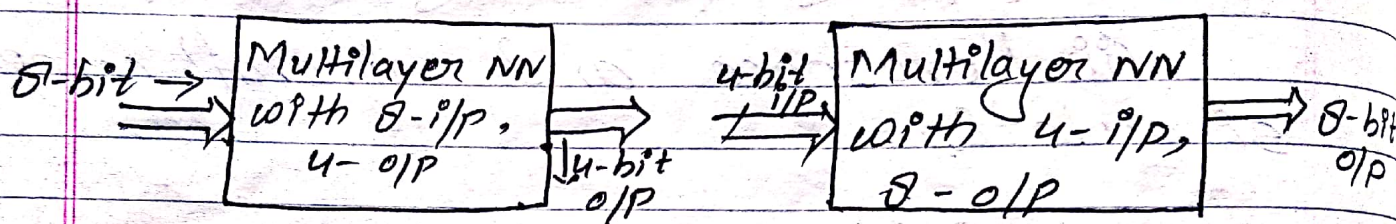
(ii) The architecture of NN is different from the architecture of microprocessor.

(iii) Requires high processing time for large neural network

(iv) Hard to setup program.

Imp # Application of NN in Image Processing

1) NN for Image Compression



(fig: block diagram of Image Compression)

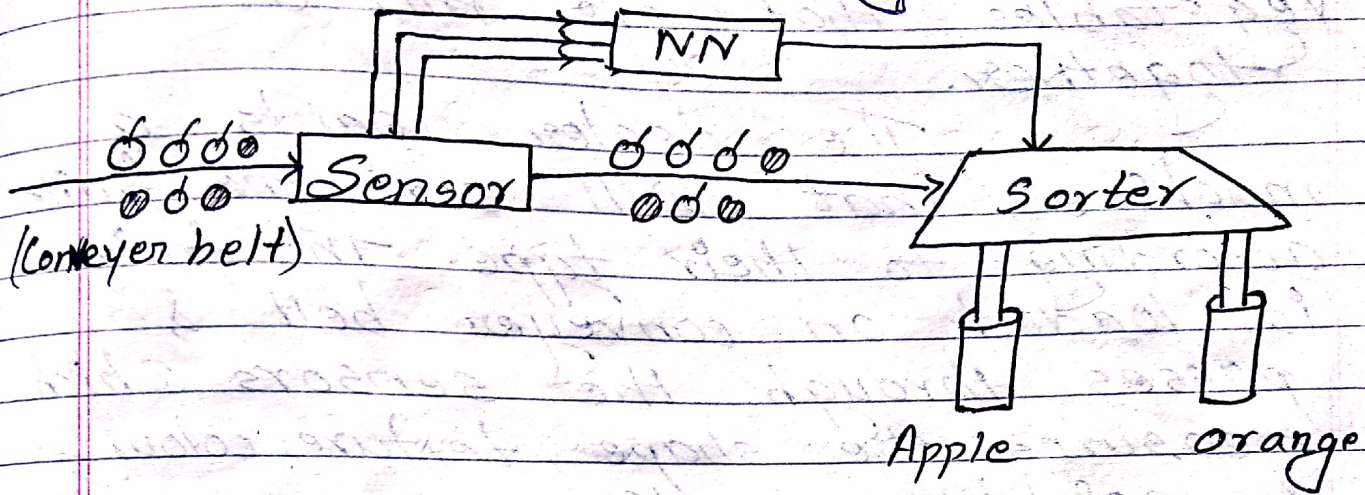
NN model have received much attention in many fields where high compression ratio is required.

Many NN approaches for image compression gives superior performance than discrete traditional approach.

Digital image generated with 8 bit may be reduced in size by ~~feed~~ feeding it to 8×4 multilayer neural network.

NN on the other side then ~~can~~ process the compressed image by 4×8 multilayer NN for reconstruction.

2) NN for Pattern Recognition



(fig: block diagram of pattern recognition (PR))

The NN is ideal tool for pattern recognition. Any recognition system needs to be trained to recognize different patterns.

NN is also a simplification of human neural network system. It is more likely to adapt the human way of solving the recognition problem than other techniques.

The design of NN system for pattern recognition starts from collecting data on each of objects that is to be recognized by the system. A dealer has a warehouse

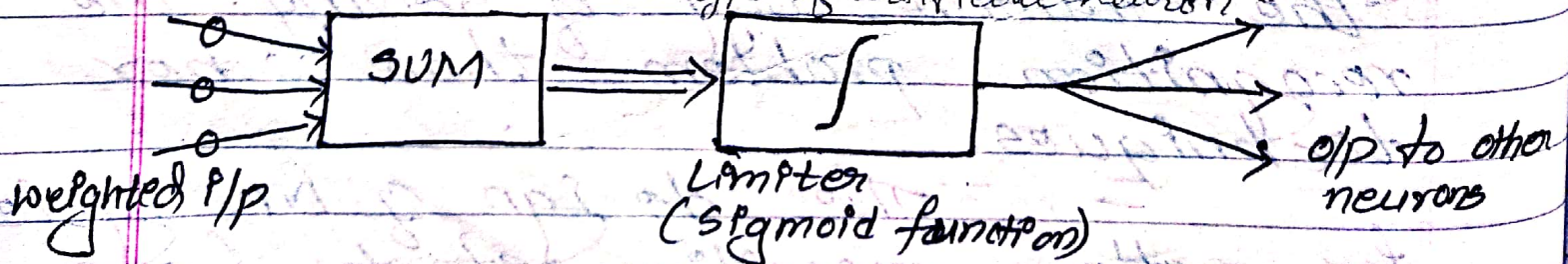
that stores a variety of fruits & vegetables that are mixed together.

The dealer wants a machine that will sort the fruits according to their type. The fruit is loaded on conveyor belt & passes through the sensors which measures the shape, texture, colour & weight properties.

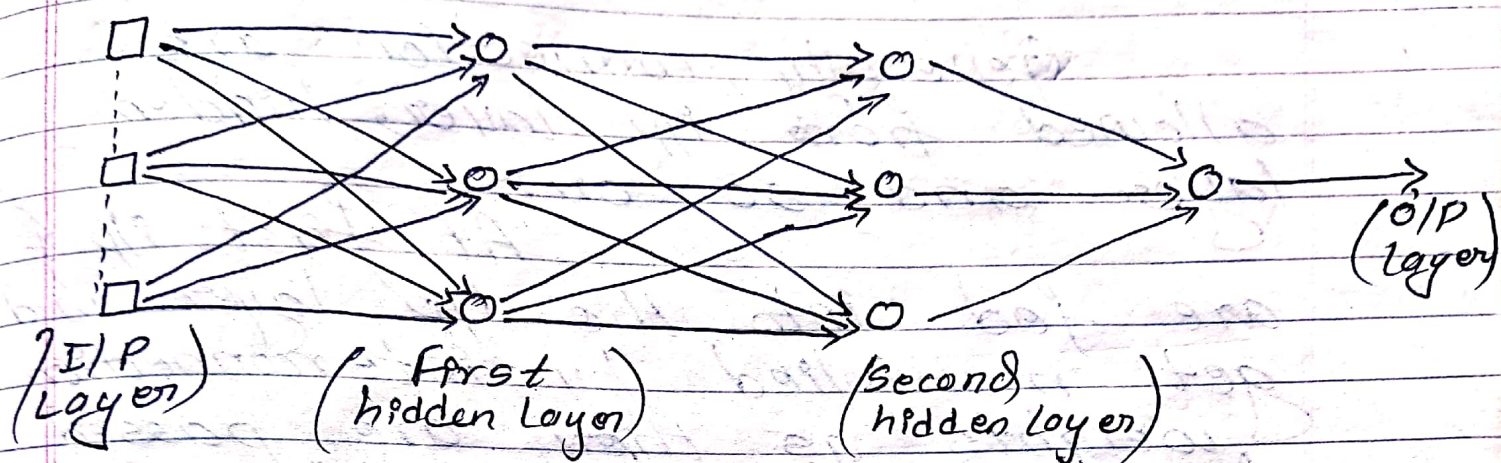
The o/p will be input to the purpose of NN ^{is to} decided what kinds of fruits are on conveyor belt.

So, fruit can be detected & directed to correct store.

3) NN for perceptron (perception) ^{or simulate}
 (computerized model to represent the ability of brain to recognize and discriminate)
 ↳ art type of artificial neuron



(fig: Block diagram of perceptron)



(fig: formation of o/p by perceptron)

It is one of the earliest neural network model. Here, inputs from one or more previous neurons are individually weighted and then summed.

The result is non linearly scaled between 0 & +1. Then, the result is passed to the neurons in the next layer for output result.

Several perceptron can be combined to form multilayer perceptron (MLP).

So, MLP is a development from the simple perceptron in which extra hidden layers are added.

Here, more than one layer can be used.

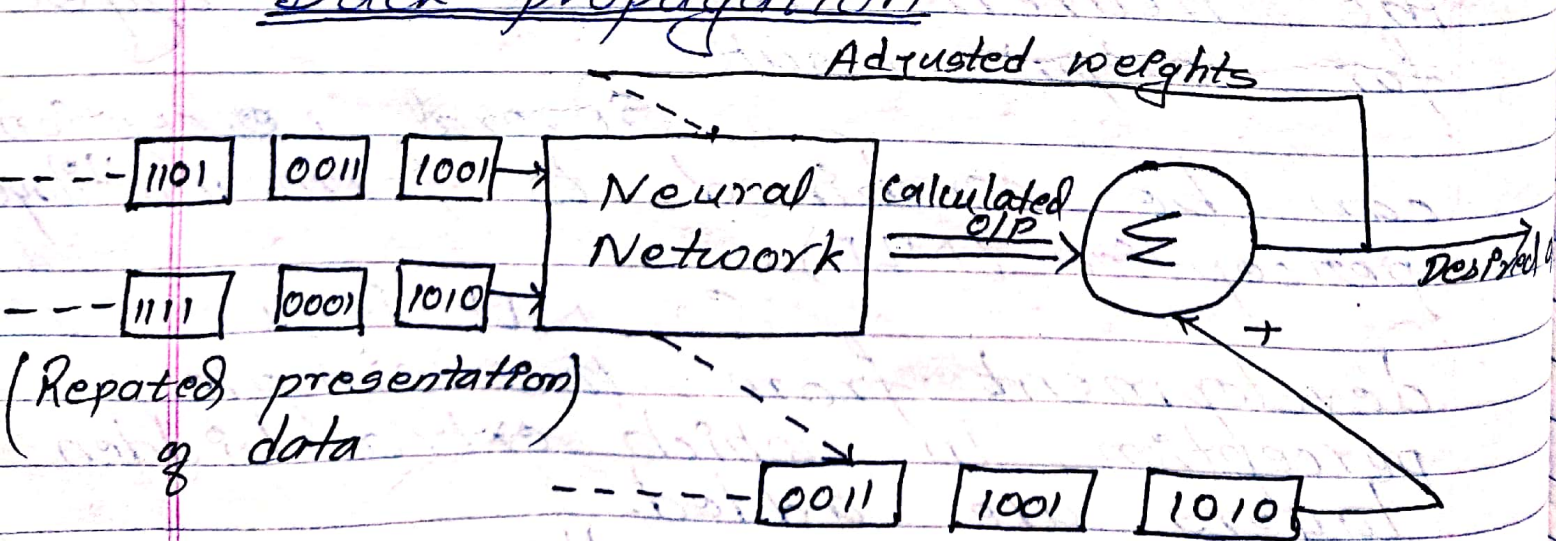
Generally, connection are allowed from i/p layer, hidden layer and so on.

At first i/p are fed into the i/p layer and get multiplied by interconnection weights as they are passed from the i/p layer to the first hidden layer.

After first layer they go the second layer & finally data is multiplied by interconnection weights & so on.

Back propagation never take the perception methods but at the time of sensing accuracy it is used.

Back propagation



(Fig: Block diagram of Back Propagation)

The MLP (multilayer Perceptron) & many other neural network can be calculated using an algorithm, this is known as back propagation.

In back propagation the i/p data is repeatedly presented to the neural network. With each presentation, the output of neural network is compared to the desired output & error is computed.

Then this error is passed as a feedback (back propagate) to the neural network.

It is used to adjust the weight such that the error decreases with each iteration. Now, the neural network calculates output and gets closer to produce the desired output.

This mechanism is known as back propagation.

Back propagation is mostly used to find the sensing accuracy which is shown in the above figure